TANH AND TAN AS RATIOS OF MEANS ON ONE PAGE

# $\tanh = \sinh \cdot \operatorname{sech}$

A half-difference times a harmonic mean, which is why speeds compose by Einstein's rule and slopes by the tangent rule.

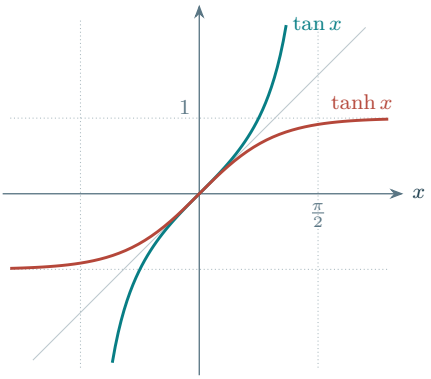
THE RULE

## A relative difference of an exponential pair

For the pair $a = \mathrm{e}^{x}$, $b = \mathrm{e}^{-x}$, with product $1$, the arithmetic mean is $\cosh x$, the harmonic mean is $\operatorname{sech} x = 1/\cosh x$, and the half-difference $\frac{a-b}{2}$, which is the arithmetic mean of $a$ and $-b$, is $\sinh x$. Their product is

$$\tanh x \;=\; \underbrace{\frac{a-b}{2}}_{\sinh x} \times \underbrace{\frac{2ab}{a+b}}_{\operatorname{sech} x} \;=\; \frac{a-b}{a+b} \;=\; \frac{\mathrm{e}^{2x}-1}{\mathrm{e}^{2x}+1}$$

the relative difference of the pair, which can never exceed $1$ in size. The imaginary pair $\mathrm{e}^{\pm \mathrm{i}x}$ gives the circular twin, $\tan x = \sin x \sec x = -\mathrm{i}\,\frac{a-b}{a+b}$, and here the denominator $a+b = 2\cos x$ can vanish, so $\tan$ runs off to infinity at $\pm\frac{\pi}{2}$.




Both leave the origin along $y = x$; $\tanh$ levels off at $\pm 1$, $\tan$ escapes.

ADDITION AND DOUBLING

## One sign separates the two addition laws

Writing $A = \mathrm{e}^{2x}$ and $B = \mathrm{e}^{2y}$ makes $\tanh x = \frac{A-1}{A+1}$, and since exponents add, $\tanh(x+y) = \frac{AB-1}{AB+1}$; clearing fractions turns this into the addition law, and the circular case follows from $\tan x = -\mathrm{i}\tanh(\mathrm{i}x)$:

$$\tanh(x \;+\; y) \;=\; \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}, \qquad \tan(x \;+\; y) \;=\; \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

Setting $y = x$ gives the double-angle forms, where the same sign decides whether the result is bounded:

$$\tanh 2x = \frac{2t}{1+t^2} \;\text{ with } t = \tanh x, \qquad \tan 2x = \frac{2t}{1-t^2} \;\text{ with } t = \tan x.$$

The first never exceeds $1$ in size for any $t$; the second blows up as $t \to 1$, the moment the doubled angle reaches a right angle.

EINSTEIN'S RULE MOTIVATED

## Doppler factors multiply, so speeds add as $\tanh$

Bondi's $k$-calculus starts from the one thing two moving observers can compare directly: light pulses sent every second arrive every $k$ seconds. Chaining two observers multiplies their $k$ factors, so $\varphi = \ln k$ adds, and the speed between them works out as the relative difference of the pair $k, 1/k$:

$$\frac{v}{c} = \frac{k - k^{-1}}{k + k^{-1}} = \tanh \varphi.$$

Multiplying $k$'s is therefore adding rapidities, and the $\tanh$ addition law is exactly Einstein's rule $u \oplus v = \frac{u+v}{1+uv/c^2}$. The double-angle form $\frac{2\beta}{1+\beta^2}$ is the speed reached by two equal boosts, and it is also the Pythagorean ratio $\frac{2pq}{p^2+q^2}$ when $\beta = \frac{p}{q}$.

| $k$ | $\varphi = \ln k$ | $v/c$ |
|---|---|---|
| $2$ | $0.693$ | $\frac{3}{5}$ |
| $2 \times 2 = 4$ | $1.386$ | $\frac{15}{17}$ |
| $3$ | $1.099$ | $\frac{4}{5}$ |
| $2 \times 3 = 6$ | $1.792$ | $\frac{35}{37}$ |

$\frac{3}{5} \oplus \frac{3}{5} = \frac{15}{17}$ and $\frac{3}{5} \oplus \frac{4}{5} = \frac{35}{37}$, both Pythagorean speeds.

WHERE THE TAN DOUBLING IS USED

## Reflections and half-angle substitution

A mirror tilted by $\theta$ turns a reflected ray by $2\theta$, so slopes double by $\frac{2t}{1-t^2}$. The substitution $t = \tan\frac{\theta}{2}$ gives $\sin\theta = \frac{2t}{1+t^2}$, $\cos\theta = \frac{1-t^2}{1+t^2}$ and $\tan\theta = \frac{2t}{1-t^2}$, turning any rational function of $\sin$ and $\cos$ into an integrable rational function of $t$, and with $t = \frac{p}{q}$ it produces the triple $(q^2 - p^2,\ 2pq,\ p^2 + q^2)$.

WHERE TANH SATURATES USEFULLY

## Bounded responses

Beyond relativity, $\tanh$ describes anything that responds in proportion at first and then saturates: the mean-field magnetisation $m = \tanh\left(\frac{Jm+h}{kT}\right)$, the dispersion of water waves $\omega^2 = gk\tanh(kh)$ between deep and shallow water, and the activation of artificial neurons, where $\tanh x = 2\sigma(2x) - 1$ is a rescaled logistic curve.

**Conventions.** $\operatorname{sech} = 1/\cosh$ and $\sec = 1/\cos$; $k$ is the relativistic Doppler factor, $\varphi$ the rapidity and $\beta = v/c$; $\sigma$ in the last box is the logistic function $1/(1+\mathrm{e}^{-x})$.